Erica Jayasundera

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Professional and Research profile

My approach is shaped by practice as much as by research. Since 2012 I have worked on security-critical banking software at National Savings Bank, Sri Lanka. That work has included supporting CBSL IT security audits and PCI-DSS compliance, which are settings where a security claim must be evidenced, reviewed by a third party, and defended under scrutiny. This is the standard I want robustness claims in machine learning to meet. I also hold an MSc in Information Security, which provides the foundation in threat modelling and systems security that I bring to adversarial machine learning.

Research Interests

- **Adversarial machine learning:** attacks, defences, and above all the methodology for evaluating them honestly
- **LLM and agent security:** prompt injection, tool misuse, and the design of defensible agent architectures
- **Security of AI-generated code:** how code produced by language models introduces, or fails to prevent, vulnerabilities
- **OSINT and influence operations:** how open-source intelligence and AI-enabled information campaigns intersect with security
- **Post-quantum cryptography migration:** the practical challenge of moving real systems, especially in finance, to quantum-resistant cryptography

Education

Years	Degree	Institution	Focus
2017 – 2019	**MSc, Cyber Security (Information Systems)** **GPA 2.60**	University of Colombo School of Computing, Sri Lanka	Vulnerability assessment and penetration testing, digital forensics, ethical hacking, security frameworks, secure software engineering
2024	**MSc, Artificial Intelligence**	University of Moratuwa, Sri Lanka	
2004 – 2008	**BSc (Hons), Software Engineering, Second Class**	Kingston University London, UK	Software Architecture & Design, Project Management, Analytical Problem-Solving
September 2022.	**Continuing professional development: Embedded Machine Learning for Edge Computing**	University of Moratuwa (ENTC),	
2025 - 2026	**Foundations of Machine Learning :**	Zuu Crew	Transitioning from data structures to model creation using classical machine learning methodologies. Python, NumPy, Pandas, scikit-learn, Matplotlib, and mathematical pillars (Algebra & Calculus).

Years	Degree	Institution	Focus
	Production ML (MLOps) :		Moving beyond standard notebooks to build, monitor, and maintain enterprise-grade ML pipelines in production. MLflow, Apache PySpark, Apache Airflow, Apache Kafka, Docker, and AWS ECS. End-to-end machine learning engineering, data orchestration at scale, automated pipeline architecture, and containerised cloud deployment.
	AI Engineer Essentials (LLMOps & Agent Systems) :		Engineering systems with Large Language Models (LLMs), production-ready multi-agent ecosystems, and cognitive architectures. LangChain, LangGraph, Hugging Face, Comet/Opik, Supabase, Qdrant, and Model Context Protocol (MCP).

All degrees were taught in English.

Research and professional experience

Senior Software Engineer, AI, Security and Financial Systems · National Savings Bank (NSB), Colombo, Sri Lanka · January 2012 – 2014 Contract

2014 – present -Permanent Staff Member.

- Contributed to information security programmes: CBSL IT security audits, PwC and EY cyber security audits, PCI-DSS compliance, and a bank-wide secure coding policy.
- Led an investigation into weaknesses in transaction anomaly detection. Produced a governance risk assessment that recommended specific control fixes.
- Built secure API integrations between core banking platforms and third-party payment gateways, with input validation, authentication controls and TLS.
- Introduced code review and automated regression testing, which cut production incidents by about 35%.
- Developed AI applications (NLP, computer vision, predictive analytics) for fraud detection and financial workflow automation.
- Led QA from 2012 to 2015: built the SQA toolkit, the regression test harness and the audit process, and mentored QA and development staff.

Co-founder and Lead AI Engineer · Mindview AI Sri Lanka (hybrid) · 2025 – present

- Built VibeSec, a vulnerability scanner for AI-generated code, and 7/24 Office, a secure multi-agent platform
- Researched attacks on LLMs, including prompt injection, jailbreaking and model extraction, feeding
- Wrote technical advisory reports on AI security, AI governance and financial fraud.
- Earlier work (2009–2011): data warehousing and predictive analytics tools, plus security QA consultancy on ISO 27001 implementation and vulnerability assessment.

Earlier roles

- Private Tutor and Freelancer on Fivver
- Software Developer : Mobile Apps, Rich Media House, London, 2008 – 2009 (internship from July 2007).
- Web Developer and Game Designer (freelance), oDesk, 2007 – 2011.

Selected projects (For the code repositories : https://github.com/ericajaya/)

Project	Years	What it is and what it showed	Methods and stack
AI SaaS Assessor	2026	Assessor2026A local web app that checks a SaaS product against a 58-page assessment guideline and produces a Word report of High, Medium and Low issues. A passive scan of a public address (transport, headers, cookies, email and domain hygiene, trust and AI-transparency pages) is combined with a 187-item checklist. An 88-item extension for autonomous AI agents switches on when the product includes them. A separate page benchmarks an agent's task completion, latency and cost.	Node.js, docx, headless Edge with axe-core, ISO 27001, SOC 2 and EU AI Act mapping
Valid Perturbations and seqpertur	2026	A dissertation thread on adversarial robustness of genomic and protein sequence models. A robustness number is only meaningful once the admissible perturbation set is defined biologically, because a Hamming budget admit an edit that truncates a protein and forbid one that leaves it unchanged. The reference implementation has 7 modules and 105 tests. Six dissertation chapters are drafted.	Python, perturbation sets, off-model validators, adaptive attacks
Agent Replay Debugger	2026	Records every model call, tool call, state change and approval of an AI Debugger agent to an append-only log, replays the run exactly, and names the step where two histories part. Runs with no dependencies and no network calls. event log, LCS-	Node.js, JSON Lines aligned run diff

Project	Years	What it is and what it showed	Methods and stack
Aegis Forge	2026	A platform where security artifacts (policy, threat model, control, detection rule, incident-response playbook, network design) must pass an automated harness test before they can be approved. The pipeline runs 68 tests, with 42 modelled threats and a fourteen-document SDLC set behind it.	TypeScript, Node.js, SQLite, React, Vite
Colophon	2026	A provenance, forensic-watermarking and protected-delivery toolkit for film, books, images, audio and research documents. Ed25519 sealing with timestamp proof, per-recipient watermarking, an encrypted vault with per-recipient key wrapping, and perceptual-hash leak patrol. 49 tests pass and it has no dependencies.	Node.js, Ed25519, RFC 3161, DCT, Tardos codes
LLM attack-surface mapping	2023 – present	A catalogue of attacks on production LLM systems (indirect prompt injection, model extraction through APIs, a jailbreak taxonomy) and defence-in-depth architectures. It feeds both my papers and VibeSec's detection rules.	Adaptive attack design, threat modelling
VibeSec, a vulnerability scanner for AI-generated code	2024 – present	A command-line tool that detects 12 OWASP-aligned weakness patterns that LLM coding assistants introduce, such as SQL injection, hardcoded credentials, insecure deserialisation and prompt-injection sinks. It runs in CI/CD pipelines.	Python, AST analysis, LangChain, GitHub Actions

Project	Years	What it is and what it showed	Methods and stack
VibeSec IDP	2026	An internal developer platform for a security consultancy. Provisioning sits behind a live policy gate, engagements get isolated environments with enforced expiry, compliance is computed from deployment descriptors, and an evidence pipeline verifies redaction. Thirteen scanners sit behind three interfaces	Next.js, TypeScript, SQLite, Terraform, Crossplane, Rego, Argo, Vault
7/24 Office, a secure autonomous multi-agent platform	2024 – present	A 3,500-line agent platform with 26 tools and a three-layer memory pipeline. It uses scoped tool permissions, agent isolation and input sanitisation against prompt injection and agent hijacking.	Python, FastAPI, LangGraph, Qdrant, PostgreSQL, Redis, MCP
Post-Quantum Cryptography Migration Guide (AUT)	2025 – present	A migration strategy for moving enterprise systems to post-quantum cryptography.	Cryptographic inventory, migration planning
NDB Sri Lanka fraud case, forensic advisory series	2024	An independent forensic analysis of the National Development Bank fraud covering governance failures, gaps in anomaly detection and weak regulatory oversight.	Financial forensics, control-gap analysis
Kapruka gift-concierge agent	2025	A multimodal shopping agent with a web crawler, persistent memory and multi-agent coordination. Delivered with an IEEE-style technical report.	Python, CLIP, Qdrant, Streamlit

Publications and books (https://cmb.academia.edu/ericajayasundera , https://www.amazon.com/stores/Erica-Jayasundera/author/B0H8M7QJGM)

Books, non-fiction

1. Jayasundera, E. (2026). *Look at the Logs: What your mind and body are trying to tell you about your work*. First edition. ISBN 979-8-16-072582-6. (available on Amazon)
2. Jayasundera, E. (2026). *The Meeting That Never Happened: Cybercrime and trust*, complete illustrated edition. MindViewAI. A case-study handbook of 51 evidence-graded cybercrime cases, including deepfake fraud, with the controls that would have stopped each one. (available on Amazon)
3. Jayasundera, E. (2026). *How Machines Know: An illustrated beginner's guide to the technologies that sense, learn, decide — and prove*. 30 chapters, 88 drawings. Status: proof.
4. Jayasundera, E. (in preparation). *From SE 2.0 to SE 5.0: A software engineer's roadmap through the agentic era*. MindviewAI. Chapters 1–5 drafted.
5. Jayasundera, E. (in preparation). *Four Books on Red Teaming, Blue Teaming, Purple Teaming and Yellow Teaming in the era of AI (Practical View)* . (will be published on amazon)
6. Jayasundera, E. *Copyrights and Sri Lankan Music*, Volumes I and II. (available on Amazon)

Books, fiction

1. Jayasundera, E. *The Cardinal Cipher*. Cryptography thriller.
2. Jayasundera, E. *The Aristotle Cipher: A novel of medieval Dublin*.
3. Jayasundera, E. *A Study in Certainty*. Forensic mystery about an AI system that never says "I do not know". (available on Amazon)
4. Jayasundera, E. *The Missing Class*. The second Erica Holmes casebook, on the failure modes of forensic classifiers. (will be on Amazon)

Research reports and proposals

1. Jayasundera, E. (2026). *Measuring What Breaks: Reliable robustness evaluation against adaptive adversaries, from image classifiers to language-model agents*. PhD dissertation proposal.
2. Jayasundera, E. (2026). *Closing the Gap: AI adoption in the European Union*. Policy research report, 54 references.
3. Jayasundera, E. (2025). *Post-Quantum Cryptography Migration Guide*.
4. Jayasundera, E. (2025). *AI Trends and Cost-Benefit Analysis: Emerging AI economy roles*. Mindview AI.
5. Jayasundera, E. (2024). *NDB Sri Lanka Fraud Case: Governance failures, anomaly detection and regulatory weaknesses*. Advisory series, VibeSec.
6. Jayasundera, E. (2024). *Post-Context and Harness Engineering AI Paradigms*. Research report, Mindview AI.
7. Jayasundera, E. (2024). *Emotionally Adaptive Agents*. Research proposal, AUT and Mindview AI.

Certifications and training

Year	Certification or programme	Issuer
2026	Taking Circular Innovation to Market	ICRES, 2nd International Conference on Resource Efficiency
2026	Generative AI Leader	Google Cloud
2024 – 2026	AI engineering bootcamp: multi-agent NL2SQL platform and multimodal AI agent, with IEEE-style technical reports	Zuu Crew AI
2022 – 2023	Digital Sherlocks (360/OS): advanced OSINT, deepfake detection, state-sponsored information operations	Atlantic Council Digital Forensic Research Lab (DFRLab)
2022 – present	CISSP (in preparation)	
Earlier	McKinsey Forward Programme; Machine Learning (Stanford, Coursera);	

Year	Certification or programme	Issuer
	Reproducible Research (Johns Hopkins); Big Data with Apache Spark; Scalable Machine Learning (edX)	

Technical skills

Area	Skills
AI and ML security	Adversarial ML, model evasion, prompt injection, LLM red-teaming, jailbreak analysis, RAG and agent security
Offensive and application security	Vulnerability assessment and penetration testing, SAST and DAST, secure code review, OWASP Top 10 and ASVS, API and supply-chain security
Threat intelligence and forensics	OSINT, influence-operation detection, STRIDE threat modelling, MITRE ATT&CK, digital and financial forensics
Governance and compliance	NIST CSF, ISO 27001, PCI-DSS, CBSL IS guidelines, security audits
Machine learning	PyTorch, Keras, Transformers, GANs, reinforcement learning, NLP, computer vision
Programming and infrastructure	Python, Java, C++, SQL, Bash, JavaScript; FastAPI, LangChain, LangGraph; Docker, GitHub Actions, AWS, Azure, GCP; PostgreSQL, Redis, Qdrant

Teaching, mentoring and languages

- **Teaching:** I run free information security sessions for university students.
- **Mentoring:** I mentored junior developers and ran secure-coding knowledge-sharing sessions at NSB, I also mentored several MSc and undergraduates for their technical skills.
- **Languages:** English, native level (all degrees taught in English). Swedish, beginner, French .